Least-Squares Filter Design

From prediction error to the normal equations

ERTH 467 — Geoscientific Signal Processing

What are we trying to achieve?

- Known input signal s_n and desired output (target) y_n.
- Design a filter f_k so that the convolution predicts the target:

$$\widehat{y}_n = \sum_{k=0}^{M-1} f_k s_{n-k}$$

Matrix representation

$$\boxed{\widehat{\mathbf{y}} = \mathbf{S}\mathbf{f}}$$

S contains shifted versions of the input signal; **f** contains the unknown coefficients.

Convolution as a matrix multiplication

For zero-padded causal convolution, a Toeplitz matrix is typical:

$$\underbrace{\begin{bmatrix} s_0 & 0 & 0 \\ s_1 & s_0 & 0 \\ s_2 & s_1 & s_0 \\ s_3 & s_2 & s_1 \\ \vdots & \vdots & \vdots \end{bmatrix}}_{\mathbf{S}} \underbrace{\begin{bmatrix} f_0 \\ f_1 \\ f_2 \end{bmatrix}}_{\mathbf{f}} = \underbrace{\begin{bmatrix} \widehat{y}_0 \\ \widehat{y}_1 \\ \widehat{y}_2 \\ \widehat{y}_3 \\ \vdots \end{bmatrix}}_{\widehat{\mathbf{y}}}$$

- $\mathbf{S}$: $N \times M$, $\mathbf{f}$: $M \times 1$, target $\mathbf{y}$: $N \times 1$.
- **Periodic convolution** gives a *circulant* $\mathbf{S}$ instead.
- The following least-squares derivation works for *either* matrix.

Define the prediction error

$$\boxed{\mathbf{e} = \mathbf{y} - \widehat{\mathbf{y}} = \mathbf{y} - \mathbf{S}\mathbf{f}}$$

- Usually we cannot match every target sample exactly.
- We therefore choose the filter that minimizes the **sum of squared errors**.

Least-squares objective: a scalar function of $\mathbf{f}$

$$\boxed{J(\mathbf{f}) = \sum_{n=0}^{N-1} e_n^2 = \mathbf{e}^T\mathbf{e} = (\mathbf{y} - \mathbf{S}\mathbf{f})^T(\mathbf{y} - \mathbf{S}\mathbf{f})}$$

How do we locate the minimum?

- In elementary calculus: set the derivative of a differentiable function to zero.
- Here J depends on M filter coefficients $f_0, \ldots, f_{M-1}$.
- Set every partial derivative to zero, or equivalently the gradient:

$$\nabla_{\mathbf{f}} J = \begin{bmatrix} \partial J/\partial f_0 \\ \partial J/\partial f_1 \\ \vdots \\ \partial J/\partial f_{M-1} \end{bmatrix} = \mathbf{0}$$

Our task

Compute $\nabla_{\mathbf{f}} J$ from the squared-error expression.

Step 1: expand the squared prediction error

$$
\begin{aligned}
J(\mathbf{f}) &= (\mathbf{y} - \mathbf{S}\mathbf{f})^T(\mathbf{y} - \mathbf{S}\mathbf{f}) \\
&= (\mathbf{y}^T - \mathbf{f}^T\mathbf{S}^T)(\mathbf{y} - \mathbf{S}\mathbf{f}) \\
&= \mathbf{y}^T\mathbf{y} - \mathbf{y}^T\mathbf{S}\mathbf{f} - \mathbf{f}^T\mathbf{S}^T\mathbf{y} + \mathbf{f}^T\mathbf{S}^T\mathbf{S}\mathbf{f}.
\end{aligned}
$$

Key observation

$\mathbf{y}^T\mathbf{S}\mathbf{f}$ and $\mathbf{f}^T\mathbf{S}^T\mathbf{y}$ are equal scalars (transposes of each other).

$$
\boxed{J(\mathbf{f}) = \mathbf{y}^T\mathbf{y} - 2\mathbf{f}^T\mathbf{S}^T\mathbf{y} + \mathbf{f}^T\mathbf{S}^T\mathbf{S}\mathbf{f}}
$$

Step 2: differentiate term by term

Only $\mathbf{f}$ varies; $\mathbf{S}$ and $\mathbf{y}$ are known.

$$\nabla_{\mathbf{f}}(\mathbf{y}^T\mathbf{y}) = \mathbf{0} \qquad \text{constant term}$$

$$\nabla_{\mathbf{f}}(-2\mathbf{f}^T\mathbf{S}^T\mathbf{y}) = -2\mathbf{S}^T\mathbf{y} \qquad \text{linear term}$$

$$\nabla_{\mathbf{f}}(\mathbf{f}^T\mathbf{S}^T\mathbf{S}\mathbf{f}) = 2\mathbf{S}^T\mathbf{S}\mathbf{f} \qquad \text{quadratic term}$$

Why a factor of 2 in the last term?

$\mathbf{S}^T\mathbf{S}$ is symmetric; for symmetric $\mathbf{A}$, $\nabla_{\mathbf{f}}(\mathbf{f}^T\mathbf{A}\mathbf{f}) = 2\mathbf{A}\mathbf{f}$.

Step 2: collect the gradient

$$\boxed{\nabla_{\mathbf{f}} J = -2\mathbf{S}^T\mathbf{y} + 2\mathbf{S}^T\mathbf{S}\mathbf{f}}$$

Factor the expression:

$$\boxed{\nabla_{\mathbf{f}} J = -2\mathbf{S}^T(\mathbf{y} - \mathbf{S}\mathbf{f}) = -2\mathbf{S}^T\mathbf{e}}$$

- Each gradient component describes the change in total squared error when one filter coefficient changes.
- We seek a filter for which **all** these derivatives vanish.

Step 3: set the gradient to zero

$$\nabla_{\mathbf{f}} J = \mathbf{0}$$

$$-2\mathbf{S}^T\mathbf{y} + 2\mathbf{S}^T\mathbf{S}\mathbf{f} = \mathbf{0}$$

$$\mathbf{S}^T\mathbf{S}\mathbf{f} = \mathbf{S}^T\mathbf{y}$$

Normal equations

$$\boxed{\mathbf{S}^T\mathbf{S}\mathbf{f} = \mathbf{S}^T\mathbf{y}}$$

A system of M linear equations for the M unknown filter coefficients.

Solving for the least-squares filter

Formal solution (when $\mathbf{S}^T\mathbf{S}$ is invertible)

$$\boxed{\mathbf{f}_{\mathrm{LS}} = (\mathbf{S}^T\mathbf{S})^{-1}\mathbf{S}^T\mathbf{y}}$$

- This is the expression shown on the lecture slide.
- **Numerically:** solve $\mathbf{S}^T\mathbf{S}\mathbf{f} = \mathbf{S}^T\mathbf{y}$; do not explicitly form the inverse.
- If $\mathbf{S}$ has dependent columns, an ordinary inverse does not exist; a pseudoinverse or regularization can be used.

Why is this stationary point a minimum?

The Hessian (matrix of second derivatives) is

$$\nabla^2_{\mathbf{f}} J = 2\mathbf{S}^T\mathbf{S}.$$

For any vector $\mathbf{v}$,

$$\mathbf{v}^T\mathbf{S}^T\mathbf{S}\mathbf{v} = \|\mathbf{S}\mathbf{v}\|_2^2 \geq 0.$$

Consequence

J is a **convex quadratic**: any solution of the normal equations is a global minimum. If $\mathbf{S}$ has independent columns, this minimum is unique.

Geometric meaning: the residual is orthogonal

At the optimal filter,

$$\nabla_{\mathbf{f}} J = -2\mathbf{S}^T \mathbf{e}_{\mathrm{LS}} = \mathbf{0}$$

so

$$\boxed{\mathbf{S}^T \mathbf{e}_{\mathrm{LS}} = \mathbf{0}}.$$

- The remaining prediction error is orthogonal to every column of $\mathbf{S}$.
- No further adjustment using the available shifted input signals can reduce the squared error.

Connection to correlation

Define

$$\mathbf{R} = \mathbf{S}^T\mathbf{S}, \qquad \mathbf{p} = \mathbf{S}^T\mathbf{y}.$$

Then the normal equations read

$$\boxed{\mathbf{R}\mathbf{f} = \mathbf{p}}.$$

- $\mathbf{R}$: inner products between shifted copies of the **input** (autocorrelation-related).
- $\mathbf{p}$: inner products between shifted input and **target** (cross-correlation-related).
- Boundary conventions (zero padding vs periodicity) affect exact entries.

Optional: derive the equations one coefficient at a time

$$J = \sum_{n=0}^{N-1} \left(y_n - \sum_{k=0}^{M-1} S_{nk} f_k \right)^2$$

By the chain rule, differentiating with respect to a single f_j gives

$$\begin{aligned} \frac{\partial J}{\partial f_j} &= 2 \sum_{n=0}^{N-1} \left(y_n - \sum_{k=0}^{M-1} S_{nk} f_k \right) (-S_{nj}) \\ &= -2 \sum_{n=0}^{N-1} S_{nj} e_n. \end{aligned}$$

Setting this equal to zero and rearranging:

$$\boxed{\sum_{n=0}^{N-1} S_{nj} \sum_{k=0}^{M-1} S_{nk} f_k = \sum_{n=0}^{N-1} S_{nj} y_n}$$

This is the jth row of $\mathbf{S}^T \mathbf{S} \mathbf{f} = \mathbf{S}^T \mathbf{y}$.

Summary: from prediction error to filter

1. Predict: $\widehat{\mathbf{y}} = \mathbf{S}\mathbf{f}$.
2. Define error: $\mathbf{e} = \mathbf{y} - \mathbf{S}\mathbf{f}$.
3. Minimize $J(\mathbf{f}) = \mathbf{e}^T\mathbf{e}$.
4. Differentiate: $\nabla_{\mathbf{f}} J = -2\mathbf{S}^T\mathbf{y} + 2\mathbf{S}^T\mathbf{S}\mathbf{f}$.
5. Set equal to zero and solve:

Main result

$$\boxed{\mathbf{S}^T\mathbf{S}\mathbf{f} = \mathbf{S}^T\mathbf{y}}$$